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MFLCP: Personalized Multimodal Federated Learning via Collaborative Prompting with Missing Modalities

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Abstract

Multimodal Federated learning (FL) is a collaborative and privacy preserving machine learning paradigm for multimodal data. With the impressive performance of large-scale pre-trained models, an increasing number of these models are being applied to FL. However, multimodal data in the real world is usually incomplete in modalities. Additionally, directly applying these large-scale pre-trained models in the federated learning framework will lead to the problem of high computational and communication costs. To address these problems, we propose a novel Personalized Multimodal Federated Learning method via Collaborative Prompting with Missing Modalities (MFLCP). Specifically, we propose an efficient large-scale pre-trained personalized multimodal federated learning framework. To address the issue of incomplete modalities in multimodal data, we propose a modal projection-aware collaborative prompting strategy for incomplete multimodal federated learning. Different categories of prompts are designed for the missing categories, and the modality mapping part and leverage complementary semantic information from different modalities are designed to guide prompt learning, promoting better interaction between modalities. In addition, we propose a communication optimization method for efficient multimodal federated learning, which reduces the parameters of multimodal pre-trained models in the process of federated communication transmission, enhances the speed of local training, and significantly improves convergence

speed by integrating large-scale pre-trained models in a lightweight manner. Meanwhile, we establish a personalized adaptive update mechanism for the federated local model, which can adaptively update the local model according to the characteristics of local data, effectively reduces the impact of data heterogeneity. Extensive experimental results on several benchmark datasets demonstrate that the proposed method outperforms the state-of-the-art baselines.

CCS Concepts

• Security and privacy; • Computing methodologies;

Keywords

Personalized multimodal federated learning; Multimodal fusion; Prompt learning; Communication optimization

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1 Introduction

In recent years, large-scale pre-trained foundational models have demonstrated state-of-the-art performance in both natural language processing (NLP) and computer vision (CV), such as language models [14, 28] and vision models [3, 8, 23, 30]. In reality, some data require privacy protection, especially in privacy-sensitive fields such as healthcare and finance. These data cannot be uploaded and involved in model training. Therefore, it is very necessary to federalize the pre-trained models.

Recent federated learning approaches for large models [24, 26, 36] mainly focus on unimodal applications such as computer vision, natural language processing, and speech recognition. [32, 38] studied the problems of modality and data heterogeneity existing in the direction of multimodal federation and achieved good results.

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However, in the real world, some multimodal data have incomplete modalities. In such cases, the performance of the model will decline. As shown in [7], missing modality issues in federated learning are a significant problem that needs to be addressed. But the existing multimodal federated methods rarely take this into account. Thus, **the first challenge** is addressing the problem of incomplete modalities in multimodal federated learning.

The application of pre-trained large models within federated learning frameworks holds great potential. The inspiring performance of these large-scale pre-trained models indicates that there are broad prospects for applying them to multimodal federated learning deployments. However, **the second challenge** in building and training these large-scale pre-trained models in federation is the high computational and communication costs, which are beyond the reach of most computing devices and data centers. Recently proposed prompting-based tuning methods can effectively solve this problem. [21, 22, 27] freeze all parameters of the pre-trained models, adjusting only the continuous prompts, thus reducing the number of parameters to be learned while achieving strong performance. In multimodal prompt-based learning, models such as PromptFuse, BlindPrompt[19], and PMF[18] effectively facilitate modality interaction and fusion across two unimodal transformers, while MMF[37] shows robust performance even in the presence of missing modalities. On the one hand, these prompts methods have not been deployed in the federated framework. On the other hand, although MMF addresses the issue of modality missing, it fails to effectively perform the interaction between modalities.

In response to the two challenges mentioned above, we propose a large-scale pre-trained personalized multimodal federated learning framework called Personalized Multimodal Federated Learning via Collaborative Prompting with Missing Modalities (MFLCP), designed to better adapt to multimodal federated learning tasks for incomplete modality, leveraging the high performance while minimizing training and communication overhead. First, we introduce a prompt-based dual-stream multimodal model, based on a parallel module structure, which easily integrates additional modalities, freezes all parameters of the pre-trained models, and adjusts only the continuous prompts. This reduces the client-side training time and the load of federated communication parameters. Second, to resolve challenge 1, we propose a prompt fusion and interaction module, where the missing type prompts are divided into three categories according to modality absence. The modal information prompts and fusion prompts aim to extract necessary information for each modality and identify the modality absence. After a non-linear mapping layer, the information of one modality is passed to another modality to achieve better cross-modal understanding. Third, to resolve the challenge 2, we only upload the trainable prompts and some parameters to reduce the number of communication parameters. Meanwhile, we introduce a local adaptive update method to update the local model, reducing the impact of data heterogeneity and making the local model better adapted to the local data distribution. By capturing the necessary information from the global model at the element level and performing personalized aggregation at the model classification head, we mitigate the impact of non-IID (Non-Independent and Identically Distributed) data distributions between clients. To evaluate the effectiveness of MFLCP, we conducted extensive experiments on two benchmark

datasets. The results demonstrate that MFLCP outperforms existing baseline methods.

Our contributions can be summarized as follows:

- (1) We propose a novel personalized multimodal federated learning framework via collaborative prompting with missing modalities (MFLCP), and establish a personalized adaptive update mechanism for the federated local model, which can adaptively update the local model according to the characteristics of local data, effectively solving the problem of data heterogeneity in multimodal federated scenarios.
- (2) We propose a modal projection-aware collaborative prompting strategy for incomplete multimodal federated learning, which optimizes the efficiency of interaction among modalities, enhances robustness in scenarios with missing modalities, and achieves stable performance even under incomplete modality conditions.
- (3) We propose a communication optimization method for efficient multimodal federated learning, which reduces the parameters of multimodal pretrained models in the process of federated communication transmission, enhances the speed of local training, and significantly improves convergence speed and overall performance by integrating large-scale pre-trained models in a lightweight manner.

2 Related Works

2.1 Prompt Tuning

Prompt tuning is applied initially in the field of neural language processing (NLP). In this approach, a “prompt” is utilized to adjust the input text, guiding the pre-trained model to perform downstream tasks. The pre-trained GPT-3 can be used by manually selecting some words before the input text[2]. To enable vision-language models to adapt to diverse vision tasks, CoOp[40] first suggests adding learnable soft prompts near input images. Then, CoCoOp[39] creates an image-conditional prompt to fully use the input features.[13] adds prompts tailored for video understanding tasks to the I-VL model, unifying various video tasks into a single framework.

When it comes to methods that utilize prompts on multimodally pre-trained models for unimodal tasks, numerous recent studies have applied prompt tuning to pre-trained vision-language models such as ViT and BERT. [11] defined fast learning as treating different cases of modality missing as different types of inputs to adapt large pre-trained multimodal models to handle scenarios with missing modalities. MMP[15] designs modality-missing-aware prompts that can be embedded into the multimodal Transformer. These prompts can improve performance under various modality-missing scenarios while reducing the need for retraining large models.

Our proposed prompt strategy is different from the above methods. Although MMP[15] can address the issue of modality missing, it simply divides prompts into three categories and employs two single-modality transformers. There is no process of modality interaction and information transfer. PMF[18] uses prompts to fuse pre-trained unimodal models for multimodal tasks, which can achieve good results with higher memory efficiency. However, it cannot perform well when modalities are missing, while real-world data often has cases of missing modalities. Our proposed prompt method

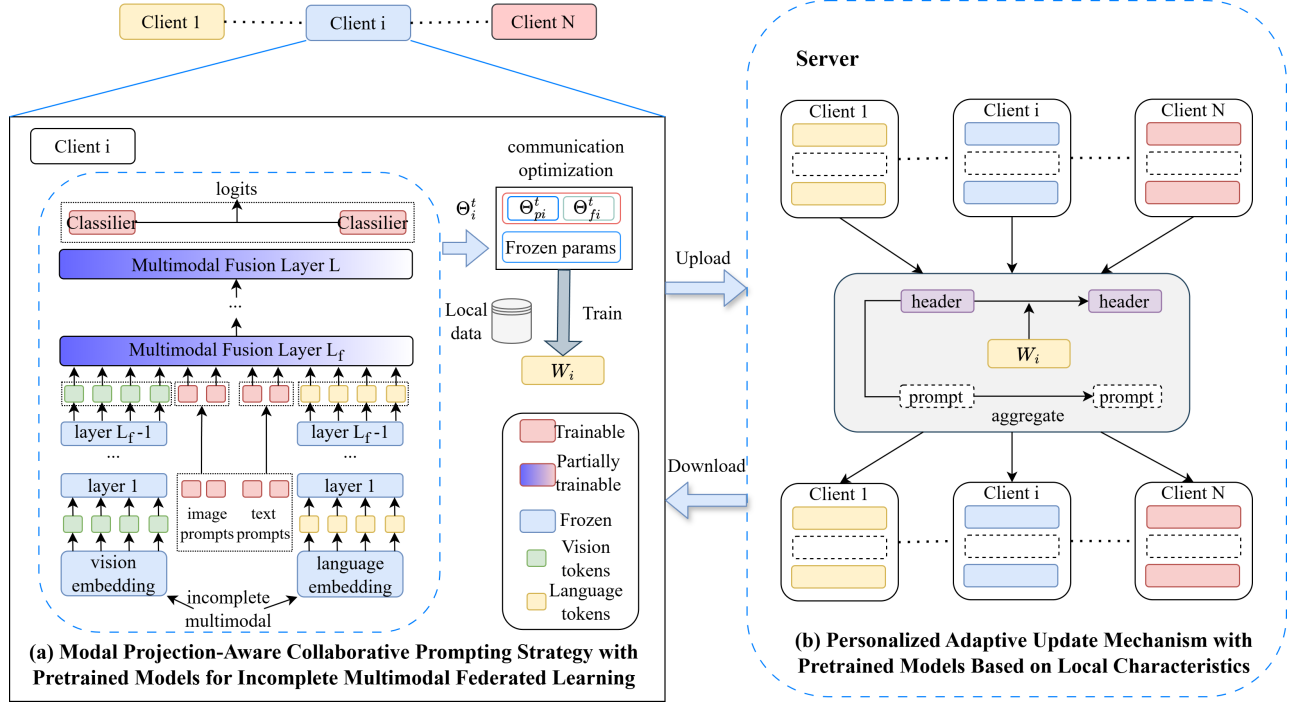


Figure 1: The overall framework of MFLCP that enhances the ability to handle missing modality data through FL by two key modules: (1) Modal Projection-Aware Collaborative Prompting Strategy with Pretrained Models for Incomplete Multimodal Federated Learning. (2) Personalized Adaptive Update Mechanism with Pretrained Models Based on Local Characteristics

enables seamless interaction and fusion between modalities while introducing tailored prompt and projection strategies to robustly handle incomplete or missing modalities.

2.2 Personalized Federated Learning

Personalized federated learning trains a model for each client using its own and others' datasets. PFEDEDIT[33] can automated model editing to optimize how the personalization layers are chosen and boost model utility, even when dealing with a wide range of data distributions, including those that are non-I.I.D. [35] proposed an alternative approach where each client collaborates only with relevant clients to achieve a more powerful model tailored to its specific objectives.[25] clusters users into different user groups and learn personalized models for each cluster by using a collaborative graph.

FedPHP[17] adopts a strategy to leverage the historical local model. It performs local aggregation between the global model and the local model. In this process, it applies a rule - based moving average and a pre - set weight (a hyperparameter) for the aggregation. FedAMP[12] allows each client to learn its own local personalized model through a message passing mechanism without constructing a global model. FedALA[34] proposed a federated learning method with adaptive local aggregation, which captures the required information from the global model of client models in personalized federated learning. When applied to multimodal data, these methods fail to consider issues such as modality interaction and modality fusion.

3 The Proposed MFLCP

3.1 Preliminary

3.1.1 Multimodal Federated Learning. Supposed we have N multimodal clients, each multimodal client k has its own private dataset $D_k = \{(x_k^{m1}, x_k^{m2})\}$, where x_k^{m1} is the image data sample in client k , x_k^{m2} is its corresponding text data. These datasets are heterogeneous (Non-IID and unbalanced). Personalized Federated Learning aim to learn individual local models $\hat{\Theta}_1, \dots, \hat{\Theta}_N$ using the dataset $D = \cup D_i (i = 1, \dots, N)$ for each client, while the data are not directly shared, which is formally described as

$$\{\hat{\Theta}_1, \dots, \hat{\Theta}_N\} = \arg \min \mathcal{G}(\mathcal{L}_1, \dots, \mathcal{L}_N) \quad (1)$$

where $\mathcal{L}_i = \mathcal{L}(\hat{\Theta}_i, D_i; \Theta)$, $\forall i \in [N]$ and $\mathcal{L}(\cdot)$ is the loss function. Θ is the global model.

3.1.2 Prompt Tuning. Given N pre-trained foundation model $\{\hat{\Theta}_1, \hat{\Theta}_2, \dots, \hat{\Theta}_N\}$ and N downstream task datasets $D_1, D_2, \dots, D_N$, prompt tuning aims to optimize a set of learnable prompt parameters $\Delta P_1, \Delta P_2, \dots, \Delta P_N$, while keeping the model parameters $\hat{\Theta}_i$ frozen. The objective is to find sets of prompts $\Delta P_1^*, \Delta P_2^*, \dots, \Delta P_N^*$, where

$$\Delta P_i^* = \arg \min_{\Delta P_i} \mathcal{L}(\hat{\Theta}_i + \Delta P_i, D_i), \forall i \in [N] \quad (2)$$

3.2 Problem Definition

For Multimodal Federated Learning using collaborative prompt-tune, the dataset used is multimodal datasets, and there are cases where some modalities are missing. The input data of the model is $D_k = \{(x_k^{m1}, x_k^{m2})\}$, when the image is missing, assign a random default value to x_k^{m1} ; when the text is missing, assign a random default value to x_k^{m2} . Then input x_k^{m1} and x_k^{m2} into the Vision Transformer (ViT) and the Language Transformer (BERT) respectively. Use Eq.(2) and Eq.(1) for local training and federated aggregation, and finally obtain the model for each client $\{\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_N\}$.

3.3 Method Overview

The overall framework of our MFLCP is shown in Figure1. We first propose three kinds of collaborative prompts for multimodal transformer, then freeze most model parameters and only train the header and learning prompt, and finally adaptively update local models after federated aggregation. MFLCP enhances downstream tasks adaptation of FL framework when the dataset is incomplete mode by two key modules: (1) Modal Projection-Aware Collaborative Prompting Strategy with Pretrained Models for Incomplete Multimodal Federated Learning; and (2) Personalized Adaptive Update Mechanism with Pretrained Models Based on Local Characteristics. We will introduce those two modules in detail as follows.

3.4 Modal Projection-Aware Collaborative Prompting Strategy with Pretrained Models for Incomplete Multimodal Federated Learning

Collaborative Prompts consist of three parts: missing type prompt (MTP), modal information prompt (MIP), and fusion prompt (FP). According to the different mode missing conditions, MTP is divided into three types: complete-MTP, text-MTP, and image-MTP, corresponding to modal integrity, missing image and missing text. The way of prompt design is shown in Fig.2.

The local multimodal fusion consists of a Vision Transformer and a Language Transformer, and performs modal fusion at the layer. The advantage of using a unimodal transformer is that modalities can be added conveniently. The image and text inputs are first processed and fed into the unimodal transformer layers to extract base features, respectively. For the sake of convenience, we define the extracted feature as z_i and z_t , which represent image feature and text feature respectively. The resulting continuous embedding is as follows:

$$z = [CLS, z_1, z_2, \dots, z_N] \quad (3)$$

Here, CLS is a special token that is added in front of the sequence. The continuous embedding z is fed into unimodal encoder. We design the starting fusion layer as L_f . The base feature extraction of each encoder can be expressed as:

$$z^{l+1} = \text{TransLayer}^l(z^l; \theta) \quad \text{if } l < L_f \quad (4)$$

where θ is the pretrained transformer parameters. Then it is subsequently passed through multimodal fusion layers, with each layer composed of three kinds of prompts, as depicted in Fig.2. The three learnable prompts are made to be intertwined with different

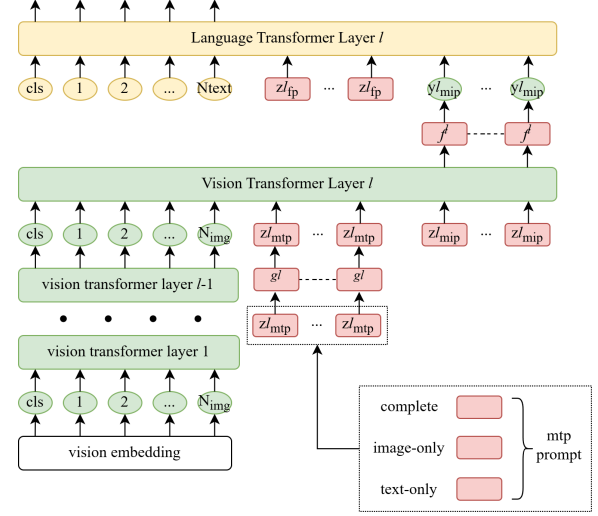


Figure 2: The collaborative prompt design

learning objectives. 'missing type prompt' (MTP) carries information about the modal absence category, and it can pass both unimodal information and absence situation to the 'modal information prompt' (MIP), then the MIP has the information from the unimodal input sequence. As for the 'fusion prompt' (FP), it performs modal fusion using the information of another modality brought by the MIP. Now, we will introduce how these three types of prompts interact with one another of each multimodal fusion layer.

First, we obtain different types of MTP respectively based on the situation of modal absence, then project the MTP into the image and text space through projection layers respectively, so as to cooperate with the features of different modalities, as:

$$\begin{aligned} Z_{mtp}^{l,T} &= \mathcal{G}^T(Z_{mtp}^l) \\ Z_{mtp}^{l,I} &= \mathcal{G}^I(Z_{mtp}^l) \end{aligned} \quad (5)$$

where $\mathcal{G}^T$ and $\mathcal{G}^I$ are the projection layers for the text modality and the image modality, respectively. The MTP contains information about common modal features and modal missing situation. The $\mathcal{G}$ can project the MTP into the image space and the text space, and we obtain $Z_{mtp}^{l,T}$ and $Z_{mtp}^{l,I}$ through the projection layers respectively. Both of them are implemented as a MLP in which the intermediate feature reduction factor is set to $r = 16$. We concatenate the corresponding MTP and MIP with the input sequence z , then it be feed to the unimodal transformer layer, which can be denoted as:

$$[\hat{z}^l || \hat{z}_{mtp}^l || \hat{z}_{mip}^l] = \text{TransLayer}^l([z^l || z_{mtp}^l || z_{mip}^l; \phi]) \quad (6)$$

where '||' is concatenation operation, and ϕ stands for the pre-trained parameters. The output of the $\hat{z}_{mtp}^l$ is extracted after the forward propagation, which has modal information and incomplete type. Then the modal information prompt $\hat{z}_{mtp}^l$ is transformed and directed towards the representation space of the other modality by utilizing a non-linear mapping function:

$$y_{mip}^l = \mathcal{D}^l(\hat{z}_{mtp}^l) \quad (7)$$

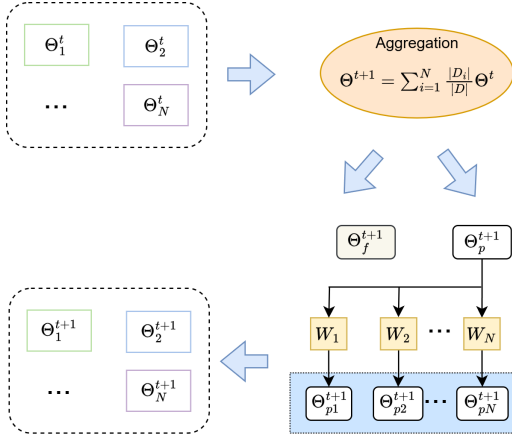


Figure 3: The Personalized Update

where $\mathcal{D}$ is a non-linear mapping function. Specifically, it consists of two linear layers with a bottleneck structure for dimension reduction, and only the first linear layer has a ReLU function. After passing through each fusion layer l , the interaction of $\hat{z}_{mip}$ in the two modal spaces can be realized.

Now, y_{mip}^l has already been projected into the space of another modality. We concatenate y_{mip}^l to the original input sequence z'^l and CP $z_{cp}'^l$ of the other modality. The concatenated sequence is input into the unimodal transformer layer of the other modality for fusing:

$$\left[\hat{z}'^{l+1} || \hat{z}_{cp}'^l || \hat{y}_{mip}^l \right] = \text{TransLayer}^l \left(\left[z'^l || z_{cp}'^l || y_{mip}^l; \phi' \right] \right) \quad (8)$$

where $[*]'$ is the another modality. $\hat{z}'^{l+1}$ and $\hat{z}_{cp}'^l$ are inputted together into the higher layer, then we take them to different linear classifiers to get the result.

Finally, we send z to the corresponding modal linear classifier, and combine the output results of the two modalities to obtain our predicted label y' :

$$y' = \frac{h_t(z'_t) + h_i(z'_i)}{2} \quad (9)$$

where h_t is the text linear classifier, h_i is the image linear classifier.

3.5 Personalized Adaptive Update Mechanism with Pretrained Models Based on Local Characteristics

For federated models with heterogeneous settings, the global model has poor generalization ability on some clients, and the global model does not necessarily conform to the modal distribution of each client. Therefore, when performing local aggregation, we use an adaptive local aggregation method, where the FedALA[34] method was referenced. However, because its method is applied to uni-modal datasets, and since we are using it for multimodal downstream tasks, we chose to personalize the two parts of the classifier, and the prompt parameters for the next training round are the aggregated global prompt. This can not only enable the client to select more suitable parameters for the local based on the category

Algorithm 1 MFLCP: Personalized Multimodal Federated Learning via Collaborative Prompting with Missing Modalities

Input: N clients, $\mathcal{L}$: loss function, Θ^0 : initial global model, a : prompt mtp, b : prompt mip, c : prompt fp
Output: Personalized local models $\hat{\Theta}_1, \dots, \hat{\Theta}_N$

- 1: **Server executes:** Initialize global model Θ_0
- 2: **for** round $t = 1$ to T **do**
- 3: **for** each client $i = 1$ to N in parallel **do**
- 4: **Adaptively update local model:**
- 5: Train the weights W_i based on local data and the local model via Eq.(10)
- 6: Adaptively receive global model based on W_i and updated model via Eq.(11)
- 7: **Local training:**
- 8: Perform modality fusion using local pre-trained encoder and learnable prompts (a, b, c) via Eq.(3) to Eq.(8)
- 9: Complete local training with loss function $\mathcal{L}$
- 10: Upload local trainable parameters to server
- 11: **end for**
- 12: **Server aggregates:**
- 13: Perform federated averaging to obtain global model Θ_t
- 14: **Server broadcasts:**
- 15: Send updated global model Θ_t to all clients
- 16: $t = t + 1$
- 17: **end for**
- 18: **Return:**
- 19: Final local model $\hat{\Theta}_i^T$

characteristics of the local data, but also enable the client model to better adapt to the distribution of the local modality after adaptive aggregation. The way of personalized update is shown in Table 3.

After the local training of the client, we upload the parameters to the server, which contains only the trainable prompts, the projection layers, and the linear classification layers, resulting in a very small amount of communication parameters. Federated averaging is carried performed on the server side to obtain the global model Θ^t . As shown in Fig. 3, the Θ^t can be split into two parts, Θ_p^t and Θ_f^t , according to the common information and personalized information. The Θ_p^t is the personalized parameters, which is made up of linear text classifier h_t and linear image classifier h_i . It will be updated adaptively by the client. The Θ_f^t is the common parameters, which is made up of learnable prompt and two projections. Since the lower layers of the model mainly contain general information, while the higher layers vary significantly according to different data distribution situations.

In the client, we initialize the value of each element in weight parameters W_i^p to one and train it based on local dataset D_i and local modal $\hat{\Theta}_i$:

$$W_i^p \leftarrow W_i^p - \mu \nabla_{W_i^p} \mathcal{L}(\hat{\Theta}_i^t, D_i^{s,t}; \Theta^{t-1}) \quad (10)$$

where $D_i^{s,t}$ is randomly sample $s\%$ of D_i , μ is the learning rate for weight learning. Then we use W_i^p and global modal Θ_{pi}^{t-1} to

adaptively aggregate local model:

$$\hat{\Theta}_{pi}^t := \Theta_{pi}^{t-1} + \left(\Theta_p^{t-1} - \Theta_{pi}^{t-1} \right) \odot W_i \quad (11)$$

where $\odot$ is a Hadamard product. And the $\hat{\Theta}_{fi}^t$ is recompletely replaced by the global parameters Θ_f^t . In this way, not only can the training parameters be kept small, but also the locally trained model can better adapt to the local data.

3.6 Training Pipeline

The overall algorithm can be seen in Algorithm 1. Concretely, the training pipeline of MFLCP mainly includes five steps:

- (1) Firstly, each client's local model is initialized with the server's global model Θ_0 .
- (2) For each client, feature extraction is carried out using the local two unimodal transformers. After that, modality fusion and interaction are performed through the Multimodal Fusion layer and the Projection layer for local training.
- (3) After all clients have completed the local updates, the learning prompts and the high-level parts of each local model are uploaded to the central server, while the other unupdated parts are retained locally.
- (4) The server receives the uploaded components and executes the aggregation operation to generate the averaged global model. Subsequently, the global model is broadcast to each distributed client.
- (5) After receiving the global model, the client partially accepts the high-level part by using the weight coefficients W_i , while the learning prompt replaces the original component, initiating the next round of learning.

By iteratively repeating Steps 2 to 5 for communication rounds T , each client can obtain a personalized local model Θ_i .

4 Experiments

4.1 Experimental Setting

4.1.1 Models and Datasets. We use two unimodal transformers as the multimodal backbone of every client, with BERT[14] as the text encoder and ViT[6] as the image encoder. All the parameters of the model are frozen, except for the image classifier, text classifier and the trainable prompts.

UPMC Food-101[29] is a multimodal classification dataset that comprises food images along with textual recipe descriptions for 101 types of food. There are a total of 90,840 image-text pairs in UPMC Food-101, and the size of different classes ranges from 790 to 956 pairs.

CrisisMMD[1] consists of 18.1k tweets that contain both paired visual and textual information. It gathers relevant tweets from seven prominent natural disasters, such as the California Wildfires in 2017. One of the objectives of this dataset is to identify the disaster's impact, including utility damage and injured or dead people.

4.1.2 Implementation Details. In the MFLCP, we set up 10 clients with 100 total communication rounds and employ the Dirichlet data partition method[10] to construct different heterogeneous

settings[20]. For local dataset, we use an ImageNet-21k[4] pre-trained vit-base model for the vision encoder and a bert-base-uncased model for the language encoder. All pretrained checkpoints are from huggingface [31]. We freeze all the parameters of both the image encoder and text encoder, and only tune the parameters of collaborative prompts and the fc layer. The prompt vectors are initialized through a Gaussian distribution(mean=0, std=0.02). In each communication round, all clients participate in local training, then uploading only the fc layer and learnable prompts for aggregation. We use SGD as the local training optimizer and set the batch size to 64.

We refer to the Dirichlet heterogeneity partition method of the FedALA[34] for data division. The ratio of the training part to the test part of the dataset is 3. Each client holds a certain proportion of training data and test data, and this part of the data is protected by privacy and not transmitted to the server. Meanwhile, in the multimodal dataset of each client, a certain proportion η of some modalities are missing. After the federated training is completed, the personalized model finally trained by each client is used for validation, and the final result is obtained by taking the average.

The downstream task of our system is the image-text multimodal classification task. For UPMC Food-101[29], we use the top-1 classification accuracy to evaluate the recognition performance. For CrisisMMD[1], we employ F1-Macro to measure the multi-label classification performance.

4.1.3 Setting of Missing Modality. We define the missing rate η as the proportion of modality-incomplete data to the train dataset. With reference to MMP[16], We stimulate three types of missing cases: missing-text, missing-image and missing-both. For missing-text case, the training data are made up of η image-only data and $1 - \eta$ modality-complete data. For missing-image case, the training data are made up of η text-only data and $1 - \eta$ modality-complete data. For missing-both case, the training data are made up of $\frac{\eta}{2}$ image-only data, $\frac{\eta}{2}$ text-only data and $1 - \eta$ modality-complete data.

4.1.4 Overall Comparisons. We compare MFLCP with method RNBT[5, 9], PMF[18], MFLCP_A, MFLCP_B. RNBT: Use ResNet50[9] as the image feature extractor and BERT[5] as the text feature extractor. PMF: Use PMF[18] as the local model. MFLCP_A: MFLCP without personalized update. MFLCP_B: MFLCP without different prompt strategies for different categories of missing modalities. PMF[18] that performs efficient and flexible multimodal fusion with prompt learning. Because this method is a non-federated model, we use FedAVG as the federated aggregation method.

We first evaluate RNBT, PMF and MFLCP methods across different missing cases including missing-image, missing-text and missing-both in the scenarios of Independent Identically Distributed (I.I.D) data distribution. Table 1 compares our method with RNBT and PMF for prompt-tuning ViT and BERT on two multimodal datasets. For UPMC Food-101[29], we set the missing rate $\eta = 0.9$ and $\alpha = 0.1$ for the heterogeneous setting. For CrisisMMD[1], we set the missing rate $\eta = 0.5$ and $\alpha = 1.0$ for the heterogeneous setting. The results are shown in Table 1.

Table 1: Overall experimental results on different methods. For UPMC Food-101 dataset, $\alpha = 0.1$, $\eta = 90\%$, the measure index is the accuracy; For CrisisMMD Dataset, $\alpha = 1.0$, $\eta = 50\%$, the measure index is the F1-Macro

Datasets	Missing Rate η	Train		Testing set				
		Image	Text	RNBT	PMF	MFLCP_A	MFLCP_B	MFLCP
Food101	90%	100%	10%	64.13	66.61	67.94	65.77	71.27
		10%	100%	79.48	83.87	84.95	88.33	87.11
		55%	55%	71.99	79.39	81.67	73.26	82.01
CrisisMMD	50%	100%	50%	38.17	41.42	38.27	41.11	45.54
		50%	100%	40.15	39.17	36.80	44.70	46.33
		75%	75%	40.82	40.65	35.90	46.20	48.12

We observe that: 1) Compared to the baseline, our method could notably promote the performance across all missing rates for various missing-modality cases. 2) Compared with missing-text and missing-both, the performance degradation caused by missing-image is smaller, which indicates that for multimodal datasets with a large amount of missing-text, we should give more consideration to taking measures to reduce performance degradation.

4.1.5 Ablation study. Component Analysis. To further validate the effectiveness of each component, we conducted ablation experiments using two versions of the proposed method. For the UPMC Food-101 dataset, set the missing rate $\eta = 90\%$ and $\alpha = 0.1$. For the CrisisMMD dataset, set the missing rate $\eta = 50\%$ and $\alpha = 1.0$. Results are shown in Table 1. Now take the results of the Food101 dataset as an example for analysis. Experiments are carried out respectively under the conditions of 90% text missing, 90% image missing, and 55% missing for each modality. In MFLCP_A, we set different types of prompts and modality information transformation layers. However, instead of adopting personalized classifier learning, we directly employed the FedAVG federated aggregation method. The corresponding three metrics were 67.94, 84.95, and 81.67, respectively. Compared with the overall metrics, these metrics decreased by 3.33, 2.16, and 0.34, respectively. These results indicate that personalizing the classifier heads for images and texts according to the local data information can effectively reduce the impact caused by data heterogeneity, making the trained model better fit the actual local situation. In MFLCP_B, we incorporated the personalized classifier, but don't set different prompts and modality information transformation layers according to the missing categories. The corresponding three metrics were 65.77, 88.33, and 73.26, respectively. Compared with the overall metrics, these metrics decreased by 5.5, -1.22, and 8.75, respectively. These results demonstrate that our use of multi-category prompts and the modality information transformation module can effectively improve the learning of modality features in the case of modality missing, especially when text is missing.

We speculate that the reason for this result is that adaptively adjusting the classifier according to the features of local data can not only reduce the impact caused by data heterogeneity when the local data is I.I.D, but also adaptively receive global parameters based on the local modality distribution in the case of a large amount of modality missing, thereby reducing the impact of modality missing.

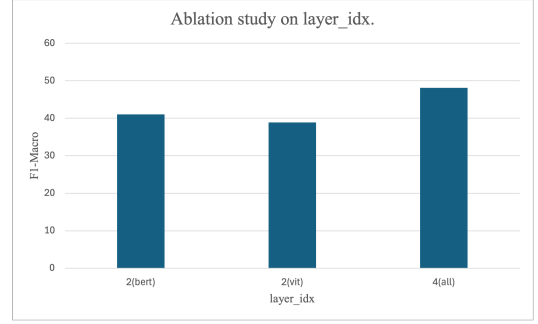


Figure 4: Ablation study on layer_idx. When layer_idx is 2(bert), only the text classification head is personalized. When layer_idx is 2 (vit), only the image classification is personalized. When layer_idx is 4 (all) with all components personalized, both image and text classification components are personalized. For CrisisMMD dataset, $\alpha = 1.0$; $\eta = 50\%$; images are 25% missing and text is 50% missing

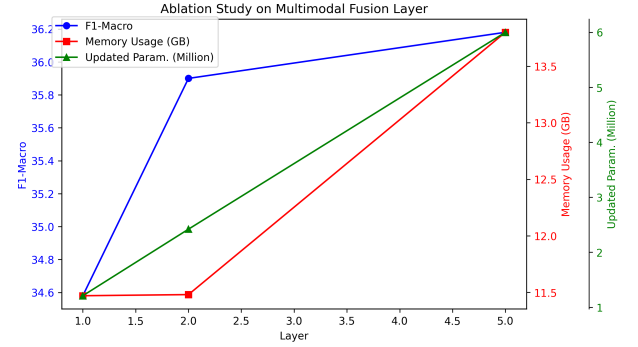


Figure 5: Ablation study on Multimodal fusion layer. For CrisisMMD dataset, $\alpha = 1.0$; $\eta = 50\%$; images are 25% missing and text is 50% missing.

4.1.6 Parameter Analysis. In this section, we conduct an analysis of the parameters in the proposed method, focusing on the number of multimodal fusion layers and the model parameters for personalized federated learning.

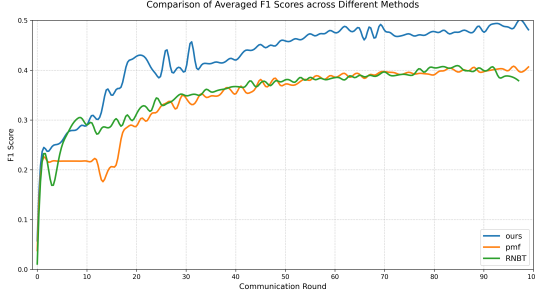


Figure 6: The Comparison of F1 Score across Three Methods

To analyze the model parameters for personalized federated learning, we conducted a total of three sets of experiments, varying the value of `layer_idx` to 2 (BERT header), 2 (ViT header), and 4 (all headers). As shown in Fig. 4, the best results are achieved when `layer_idx` is set to 4. The parameter analysis indicates that the personalization of both modal classification heads has a positive effect on model training. Furthermore, it is observed that the personalized text classification head performs better than the personalized image classification head when 50% of each modality is missing. This suggests that the data heterogeneity of text has a greater impact on federated training than that of images.

To analyze the layer parameters for multimodal prompt learning, we conducted a total of three sets of experiments, varying the value of `layer` to 1, 2 and 5. The federated aggregation model is FedAVG. The results shown in Fig. 5. First, we observe that as the number of layers increases, the interaction between modalities strengthens, enabling each modality to more effectively learn the features and missing information of the other modality. Notably, the performance improvement when increasing the number of layers from 1 to 2 is more significant than the improvement observed when the number of layers is increased from 2 to 5. Additionally, as the number of layers increases, both the memory consumption during training and the communication overhead due to parameter exchange also rise. Therefore, after carefully considering the trade-off between modality interaction and communication overhead, we select the configuration with 2 layers. This choice strikes a balance by ensuring effective modality interaction while preventing excessive communication costs in federated training.

4.1.7 Federated Convergence Analysis. We extracted the data of various indicators corresponding to different communication rounds for the three methods, RNBT, PMF, and MFLCP, under the Crisis-MMD dataset, which set $\alpha = 1.0$, $\eta = 50\%$, image-missing is 25%, text-missing is 25%. Based on this data, we plotted a curve graph, and the details are shown in Fig. 6.

We can observe that throughout the communication rounds, MFLCP achieves the highest F1 score, indicating better results compared to other methods. MFLCP shows a significant jump in the F1 score during the initial rounds, suggesting a faster initial convergence rate. In contrast, the improvement of PMF and RNBT is slower and more gradual, especially in the later rounds, which indicates that they may require more rounds or more sophisticated approaches to reach higher performance levels. Overall, the results

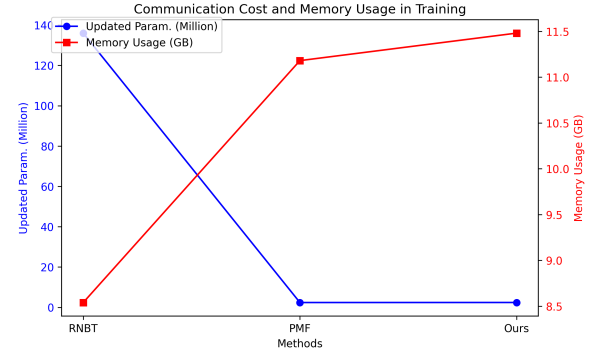


Figure 7: Communication Cost and Memory Usage in training

demonstrate that our method outperforms the other two methods in terms of both convergence speed and final performance.

4.1.8 Communication Cost. We have statistically analyzed the sizes of the updated parameters and the memory occupation under different models, and the results are shown in Fig. 7. It can be observed that when the model is training without using prompt learning but trained with ResNet and Bert instead, the number of parameters to be updated is extremely large. As a result, the communication volume during federated aggregation will be very high. However, when the fine-tuning method is employed, the number of parameters to be updated is reduced to 2.4. Meanwhile, compared with the multi-modal model PMF[18], we have designed different processing methods for the situation of modality missing. At the same time, we have added a modality mapping module. Without a significant change in the number of updated parameters and memory usage, the model can effectively deal with the situation of modality missing in multi-modal data in practical scenarios.

5 Conclusion

This paper proposes Multimodal Federated Learning via Collaborative Prompting with Missing Modalities(MFLCP), a novel large-scale pre-trained personalized multimodal federated learning framework for incomplete modality. MFLCP applies pre-trained models to federated learning while maintaining the lightweight nature of training and communication parameters, and address a critical challenge related to missing modality by proposing a module for prompt learning and adaptively aggregation. Experimental results across two datasets and three scenarios of modality missing showcase the effectiveness of MFLCP. In the future, we aim to explore the incorporation of more modalities and more parameter-efficient prompt learning for tackling more complex downstream tasks. In subsequent research, we will improve MFLCP from two aspects: 1) designing a less memory usage prompt tuning method. 2) extending MFLCP to more modalities.

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